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1 introduction

My work uses digital tools and methodologies to study language, especially language related to gender, sex, and sexuality. While my formal training is in English Literature, with subfocuses in Queer Studies and the Digital Humanities (DH), my work is at the intersection of the Humanities and Computer Science disciplines, including fields like Critical AI Studies and Computational Linguistics.

Since beginning my position at the Pratt Institute School of Information in Fall 2024, I pursue a research agenda based on three main goals:

- (1) to explore and critique computational mechanisms as heuristics for cultural phenomena;
- (2) to interrogate the contradictions and investments in present day, social understandings of gender, sex, and sexuality; and
- (3) to communicate across the disciplines.

Toward that end, I have written five papers, two of which have been published in peer-reviewed venues, and three which are currently being considered for publication at peer-reviewed venues. In addition, I have also secured an academic book contract, with the complete manuscript due in spring 2027.

In what follows, I explain the details of this work and how they fit into the goals described above.

1.1 interrogating present day contradictions in gender, sex, and sexuality

My research takes contemporary discourses on gender, sex, and sexuality as its primary subject. The aim of my research is to disentangle some of the contradictions that constitute these definitions according to the current debates, with the goal of finding middle ground between polarized views.

As of the past several years, the discourse on gender in particular is highly polarized, with various waves of reactionary gender movements, including "gender-critical feminism", "TERFism", among others. These initiatives are largely driven by conservative arguments opposing "gender ideology" or "gender theory," and grounding gender in "biological" sex.

My work examines the apparent poles of this discourse to open a space of connection, a middle ground between both sides. The idea is not to resolve or settle the disagreements between them, but to demonstrate that there are spaces of solidarity even between highly contrasting views. It suggests the idea that gender ought to be defined as an expressive phenomenon, which counters some more progressive views of gender while supporting Trans Studies theorizing on the subject. Gender as an

I have pursued this goal in two recent papers, both which are currently being considered at peer-reviewed venues.

The first paper, "What LLMs Can't See About Gender," currently under review at *FAccT 2027*, examines Natural Language Processing (NLP) methods for detecting and evaluating gender bias in language. The paper argues that these methods, which use Large Language Models (LLMs) to detect and evaluate gender bias, tend to define gender as an internal phenomenon, as interior, "personal" feelings of identification, which is associated with a progressive view of gender identity.¹ This paper critiques this practice, demonstrating that it limits what LLM bias detection and evaluation methods are able to perceive. Rather, it recommends developing methods that attend to gender as an expressive and behavioral phenomenon, drawing from recent theorizing in Trans Studies as an example. This view is more in line with conservative emphasis on gender as a physical phenomenon, though without the ultimately fascist interests in establishing biological boundaries and requirements.

The second paper, "'My Body Feels Like it's Coming Off': on Cis Embodiment in *Love Is Blind*," currently under review at *TSQ: Trans Studies Quarterly*, examines what cisgender experiences of embodiment can lend to Trans Studies understanding of body dysphoria. It analyzes a dating reality TV show, *Love Is Blind*, which places its participants in a dating experiment that prevents them from seeing their partner until they have agreed to get married. The paper argues that the dating environment, where visual access to the beloved is suppressed, simultaneously amplifies the participants' relationships to their other bodily senses, particularly the sense of touch. This intensification of bodily sensation is disrupted when the participants

¹ Jack Drescher. 2025. What is Gender Dysphoria? *American Psychiatric Association*.

finally meet in person, and many of them cannot transplant their visceral emotional connection into a physical relationship. The paper argues that this experience, although within a firmly cisgender and normative context, evokes aspects of Trans Studies theorizing of dysphoria. It therefore opens a vector of commonality between actually divergent life experiences.

1.2 reading computational mechanism as a heuristic for cultural norms

This research theme is the main argument of my forthcoming book, *Queer Code: Re-working Language Technologies*, which is in contract with Bloomsbury Academic Press.

The book explores how language technologies reduce language complexity to make it computable. Words are removed from their original context, cleaned and standardized, so that they can be stored and analyzed at scale. These increasingly automated processes encode dominant logics and extractive incentives onto language data, creating well-documented issues with data sovereignty, privacy, and most recently, bias, as current AI tools demonstrate.

Each of the book's five chapters considers how these processes of reduction operate through different language technologies, such as web scraping or text analysis. Each chapter identifies a mechanism in each technology that reduces or flattens language meaning. These mechanisms, which the book calls "technical constraints," are specific to the technology in question—for example, web scraping exhibits the technical constraint of protocol, while text analysis exhibits the technical constraint of iteration. Then, drawing from Queer Studies and related fields, the book identifies conceptual tools that re-work these constraints into new analytical methodologies to open up novel readings of language data.

The table below outlines the technology, constraint, and conceptual tool across each of the book's five chapters.

Chapter	Technology	Technical Constraint	Conceptual Tool
1. Text Capture	Web Scraping	Protocol	Reparative Reading (Sedgwick)
2. Text Analysis	Quantitative Text Analysis	Iteration	Gender Performativity (Butler)
3. Text Preservation	Markup Language	Containment	Dis-Identification (Muñoz)
4. Text Display	Screen Display	Abstraction	The Flesh (Spillers)
5. Text Generation	Machine Learning	Aggregation	Body Dysphoria (Prosser)

For example, chapter 2, "Text Analysis," explores the technology of *quantitative text analysis* and its constraint of *iteration*: a programmatic logic of

repetition in which text analysis programs execute the same action across large collections of text, one word at a time. This constraint cleans and regularizes texts en masse, with the attendant effect of suppressing minor details in language forms in favor of more frequent textual patterns. The chapter then takes Judith Butler's concept of *Gender Performativity* as a conceptual tool to re-work iteration into a new text analysis practice for studying gender in language.

In addition, my article, "Re-encoding Dominance: Queer Approaches To TEI Markup/", published in the book collection, *Digital Editing & Publishing in the 21st Century*, explores the language technology, TEI (Text Encoding Initiative), a markup standard for encoding text-based cultural heritage materials. It finds that TEI imposes a constraint of "containment," where all elements must be bounded and fixed as static elements within an ordered hierarchy. The article thinks through how this technical constraint reflects other power structures that operate upon a text, like those implicit in editorial practices, literary standards, and censorship.

1.3 communicating across the disciplines

Disciplinary and field boundaries are useful for organizing knowledge, education, and training. But the fact is that most research projects cross these boundaries in some way. In my research, however, boundary-crossing is more than a fact, it is an aspiration: to engage knowledges that otherwise may not come into contact with one another.

For my work, this means bringing knowledge from one academic domain into another. In the past two years, I have written three papers (one published and two under review) for primarily Computer Science fields like Computational Linguistics and Machine Learning/AI Ethics. In turn, I have written two papers (one published, one under review) for the Humanities fields of Trans Studies and Digital Scholarly Editing.

On the side of Computer Science, my papers include, "Some Myths About Bias: A Queer Studies Reading Of Gender Bias In NLP," published in the *Association for Computational Linguistics 2025 Proceedings*, "What LLMs Can't See About Gender," currently under review at *FAccT 2027*, and "Small Language Models and the Synthetic Middle," currently under review at *NeurIPS 2026*. These papers draw from theorizing in Queer Studies and Trans Studies in order to make recommendations to CS fields. For example, "Some Myths About Bias" paper argues that bias mitigation techniques may perpetuate certain myths about gender bias: first, that bias is a categorical value, where detection is equivalent to evaluation and determination of

harm; second, that bias is zero-sum, that it can be "deducted" language in a way that achieves equality between the genders. Rather, the paper draws from Queer Studies theorizing to argue that bias is embedded in language in ways that require more careful and thorough handling, without falling into the trap of binary thinking. Additionally, my paper "Whatt LLMs Can't See About Gender," shows how language systems will learn to discriminate against genders that are not explicitly marked as such. Drawing from the Humanities field of Trans Studies, the paper points out that theorizing gender as an internal identity harms those whose gender presentation is most visibly non-normative: the paper explains that when gender is defined as internal, genders which are apparently normative will be read as "default", while non-normative genders become hyper-visible and therefore vulnerable to stigmatization. Finally, my paper, "Small Language Models," under review at *NeurIPS 2026*, is motivated by an interdisciplinary goal: to incorporate the humanities analytical method of "close reading" into a larger quantitative experiment. This paper uses close reading to develop a novel methodology that uses Small Language Models (SLMs) to study polarized discourses on gender and transgender rights in the USA.

On the side of the Humanities, I have written two papers, one in the field of Digital Scholarly Editing and another in Trans Studies. The former paper, "Re-encoding dominance," considers how Queer Studies's concepts might help to rethink the rigidity of TEI, a text encoding language for preserving and displaying cultural and literary material in electronic formats. It considers how archival work in the history of slavery offers lessons for working within rigid encoding systems like the TEI—strategies like narrativizing between the gaps of what, in Saidiya Harman's words, "cannot be verified" (Hartman, "Venus In Two Acts", 12). The second paper, "My Body Feels Like It's Coming Off," applies the humanist methodology of close reading to results from a machine learning experiment. For the paper, I fine-tuned a Small Language Model (SLM) on transcripts from the reality TV dating show, *Love Is Blind*. Then I apply a framework from Trans Studeis to read and analyzes the model outputs. The goal of this paper is to incorporate computer science methods into a humanist methodology of close reading in a way that is legible to those trained in the humanities.

1.4 bank

While my formal training is in Literary Studies, Queer Studies, and the Digital Humanities, I am a self-taught programmer who takes computational tools both as a methodology for analyzing language. Historically, this has

meant applying a critical Humanist perspective to analyze how computational tools, like text analysis and text encoding for example, transform literary language into computable data. I was interested in what happens to language meaning, conveyed in formal structures and devices, like narrative form or metaphor, when it is digitized, computed, and stored in electronic formats.

Although *Queer Code* is written for Digital Humanities (DH) and particularly Queer DH, its scaffolded approach to technology also aims at nontechnical readerships from the Humanities, particularly from Queer Studies and Literary Studies. First, *Counter-Code* will appeal to readers from DH and Queer DH who are interested in learning how theoretical frameworks like Queer Studies (including the debates and trajectories within the field) apply to digital work. Second, the book's focus on Queer Studies and Literary Studies will appeal to readers from those backgrounds, who may be nontechnical although interested in applying close reading methodologies to digital contexts. The book's scaffolded approach to language technology as a stack of increasingly complex mechanisms, from text capture to text generation, make technical processes intelligible to readers without formal training.